Aakarsh Jha

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Education

- **Manipal Institute of Technology** — Manipal, Karnataka
  Bachelor of Technology in Computer Science — *2027*
  Relevant Coursework:
  - **DSA, OOP, DBMS, Operating Systems, Computer Networks, SDLC, Computer Organization Architecture & Embedded Systems , Digital Marketing**
- **U.S.M. Senior Secondary Public School** — Nangloi, Delhi
  Senior Secondary (10+2) — *2023*

Technical Skills

- **Languages**: C/C++, HTML, CSS, Java, SQL, PL/SQL, JavaScript, ARM
- **Web, Frameworks & Technologies**: Tailwind CSS, Bootstrap, Django, MPI, JavaFX
- **Tools & Platforms**: Git, GitHub, Linux (Ubuntu, Debian) , MySQL, Supabase, Vercel, VirtualBox, Cisco Packet Tracer, KEIL uVision, MS Excel

Positions of Responsibility

- **Management Committee Member** at Information Security Team (MIST) – practiced security through **30+** CTF writeups (TryHackMe, picoCTF, OverTheWire), TCCAM (Consultancy), and ESOM (Economics and Finance)
- Contributed to Magnify, TCCAM's consulting event applying **AI/ML** and digital transformation frameworks to real-world business problems..

Projects

- **LLM-Based DevSecOps Platform for Proactive Security Analysis**
  LLMs, GitHub Actions, CI/CD, Static Analysis, DevSecOps — *2025*
  - Architected an LLM-based DevSecOps pipeline integrated with GitHub CI/CD for near real-time security analysis **(30s per commit)**.
  - Implemented commit-diff vulnerability detection by filtering rule-based static analysis results through an LLM semantic reasoning layer to isolate SQLi and XSS flaws.
  - Automated security reports (JSON & Markdown) with feedback directly on GitHub pull requests, reducing manual review effort.
  - Designed a quality gate to block merges for high/critical vulnerabilities and ensure secure deployments.
- **Comparative Analysis of NGINX Ingress Load Balancing on Azure AKS**
  Kubernetes (AKS), NGINX Ingress, Locust, Docker — *2026*
  - Engineered a reproducible cloud testbed deploying containerized microservices on a 2-node Azure Kubernetes Service (AKS) cluster utilizing the NGINX Ingress Controller.
  - Evaluated Round Robin, Least Connections (EWMA), and IP Hashing algorithms under asymmetric traffic loads of up to 200 concurrent users via the Locust load-testing framework.
  - Discovered that dynamic EWMA and IP Hashing reduced tail-latency (P99) by up to **55%** (110–120ms) compared to the default Round Robin strategy (170ms) under baseline conditions.
  - Automated a Python data pipeline to quantify CPU distribution fairness via Coefficient of Variation (CV), revealing severe imbalance in IP Hashing (**CV=95.5%**) vs. near-perfect balance in Round Robin (**CV=0.1%**).
  - Simulated catastrophic node failure via live pod termination, establishing that NGINX's passive health checks maintained **99.99%+** availability with zero request loss during the 5.1-second Kubernetes self-healing recovery window.
- **Manipal Airlines — Airline Reservation System**
  Django, Python, JavaScript, SQLite — *2025*
  - Developed a full-stack airline reservation system with flight search, real-time seat selection, and end-to-end booking lifecycle management.
  - Engineered geospatial flight route visualization using Leaflet.js and a price trend analysis module with Chart.js, providing 7-day fare predictions and booking recommendations.
  - Designed a custom CSS design system with **50+** tokenized variables delivering a premium editorial UI with responsive layouts, skeleton loaders, and micro-animations.
- **Course Projects**
  - **Network Packet Analyzer (PCAP Project)**
    Programmed a packet capture and network traffic analysis tool using C and the PCAP library in Linux.
  - **Smart Healthcare Management System (SHMS)**
    Constructed a JavaFX-based healthcare management system to manage patients, doctors, and appointments utilizing a file-based storage architecture.

Certifications

- **Coursera**: Digital Marketing – Illinois Urbana, Google AI
- **Competitions Participated**: CreaTech 2024 (L&T), Adobe India, EY Techathon 6.0, NationBuilding Case Study, The Fifth Move 2025, Prod-A-Thon '25